



CONTACT



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FernandoVelillaHurtado



fvelillah



Turin, Italy

EDUCATION

Mechatronics Engineering
Universidad EIA - Medellin, Colombia
2021 - 2025 | Final Grade: 4.5/5

MSc Computer Engineering
Politecnico di Torino - Turin, Italy
2025 - 2027

SKILLS

Programming:

- C/C++/Python
- PLC (Ladder logic & GRAFCET)
- Assembly (in ARM and RISC-V)

Robotics & Control:

- ROS2 and MicroROS
- MATLAB/Simulink
- LabView
- Automation Studio

Embedded & CAD Design:

- Autodesk Inventor
- FreeRTOS and Arduino
- AutoCAD

AI/Software:

- Computer vision
- PyTorch
- CUDA
- LangGraph/RAG/Qdrant

Languages:

- Spanish (Native)
- English (IELTS Score: 7.5)

Fernando Velilla Hurtado

MSc Computer Engineering Student

SUMMARY

MSc Computer Engineering student specializing in Automation and Intelligent Cyber-Physical Systems, with a strong background in Mechatronics Engineering, focusing on robotics, embedded systems and automatic control. Experienced in integrating mechanical design, electronics and intelligent control algorithms to build real-world automated systems. Passionate about developing AI-driven robotic solutions that bridge the physical and digital worlds to improve efficiency, safety, and quality of life.

WORK EXPERIENCE

Maintenance Analyst

Dec 2024 - May 2025

Robot.com

I joined [Robot.com](#) maintenance team (formerly Kiwibot), where I successfully carried out repairing tasks, ensured the continued operation of a fleet of 30+ last-mile delivery robots, and coordinated the installation of charging stations for the robots in the University of Southern Indiana, located in Evansville, Indiana.

Engineering Intern

Nov 2023 - Feb 2024

Vidycom S.A.S

As an intern at Vidycom S.A.S and in collaboration with Ecolair, a company that produces technologies to improve air quality in Medellín, I worked in the development of systems for air quality measurement and control.

PROJECTS

SCARA robotic arm

2024

Developed a 3-DOF (RR-P) robotic arm for Industrial Robotics course, implementing trajectory planning for the end effector. Link to project description: [scara_project](#)

Autonomous vehicle with ROS2

2024

Developed a navigation stack in ROS2 and low-level firmware in Micro-ROS for a differential drive robot within the ASIMOV research group.

ROV Prototype

2024

Designed a tethered submersible for marine monitoring, featuring thrusters for maneuvering, a temperature sensor for data collection, and a cabled control system with a computer-based UI.

Air Quality Control

2025

Developed a high-fidelity simulation in the environment Automation Studio to validate PLC logic and firmware for a novel air purification system, implementing a Hardware in the Loop (HIL) methodology, ensuring control sequences and safety protocols in a virtual environment before deployment.

Unsupervised Anomaly Detection with Pytorch

2025

Developed and trained a Convolutional Autoencoder (CAE) for detection and localization of anomalous regions in industrial images using Pytorch. Utilized the MVTec AD benchmark reaching performance level near state-of-the-art for industrial quality control with a pixel wise ROC-AUC score of 0.83. Link to GitHub repo: [Anomaly-Detection-with-Autoencoders](#)

Multi-Agent AI Framework for Industrial Fleet Management

2026

Built a multi-agent AI solution for the company AROL (manufacturer of capping machines based in Italy) involving the full-stack development of an application to solve troubleshooting issues of customers with machines directly through a chatbot that featured the use of local models, LangGraph based harness and a Qdrant RAG engine achieving >90% success rate in benchmarks to test the agents capabilities. Link to GitHub repo: [SDPArolProject](#)



EXAMS

MSc Computer Engineering

Track in Automation and Intelligent
Cyberphysical Systems

Politecnico di Torino - Turin, Italy
2025 - 2027

EXAMINATIONS PASSED

• Computer Architectures (2026)	Grade: 22 ECTS: 10
• System and Device Programming (2026)	Grade: 30 ECTS: 8
• Data Science and Database Technology (2026)	Grade: 29 ECTS: 8
• Convex Optimization and Engineering Applications (2026)	Grade: 24 ECTS: 6
• Machine Learning for Vision and Multimedia (2026)	Grade: 28 ECTS: 6
• Modeling and Control of Cyberphysical Systems (2026)	Grade: 30 ECTS: 6
• Nonlinear Control and Aerospace Applications (2026)	Grade: 27 ECTS: 6
• Lingua Italiana II Livello	Superato ECTS: 6

Weighted avg. of grades: 27/30

Avg. grade for admission to the Final Examination: 99.15/110

FUTURE EXAMINATIONS

• Sensors, embedded systems and algorithms for Service Robotics (2026)	ECTS: 6
• Network Dynamics and Learning (2026)	ECTS: 6
• Robot Learning (2026)	ECTS: 6
• Robotic Systems (2026)	ECTS: 8
• Automotive Control Systems (2026)	ECTS: 6
• Internship (2026-27)	ECTS: 12
• Thesis (2026-27)	ECTS: 30
• Estimation, Filtering, and System Identification (2027)	ECTS: 6